

Axioms of Quantum Mechanics and the Arithmetic Structure of the Bost–Connes System

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Abstract

The five axioms of von Neumann quantum mechanics are each confirmed by experiment, yet they are not described within a single mathematical framework. Starting from the Bost–Connes system, we perform the p -adic reduction at a prime p and construct the state space

$$\mathcal{H}_p = L^2(\mathbb{Z}/p^K\mathbb{Z}), \quad K = p(p-1) - 1,$$

and argue that its structure naturally describes these five axioms. (I) C^* -positivity gives the Hilbert space through the GNS construction, (II) invariance under Galois conjugation gives the self-adjoint observables, (III) the CRT decomposition gives the tensor product, and (IV) $H = \log n$ generates the unique unitary time evolution. (V) The Born rule comes from the modular structure $S = J\Delta^{1/2}$ of the KMS state, its probability weights being the squared moduli of the values of Dirichlet L -functions on the critical line.

Keywords: Bost–Connes system; axioms of quantum mechanics; KMS state; Born rule; p -adic reduction; Dirichlet L -functions.

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§1 The Bost–Connes System

The axiomatic framework of quantum mechanics due to von Neumann [29] comprises five axioms (the formulation of Dirac [30] provides a complementary perspective): (I) the state space is a Hilbert space; (II) observables are self-adjoint operators; (III) the state space of a composite system is a tensor product; (IV) time evolution is a unitary group; (V) measurement probabilities are given by the Born rule. These five are the foundation of quantum mechanics, each of them confirmed by experiment, but they are not described within a single mathematical framework.

The experiments that establish them are of many kinds: black-body radiation, the photoelectric effect, atomic spectra, interference, the statistics of composite systems. What these experiments exhibit in common is only three properties: excitations are countable, energies add, independent composition multiplies.

The Bost–Connes system [1, 2, 3, 4, 5, 6, 7, 8, 9, 10] is a quantum statistical mechanical system constructed in 1995,

$$\mathcal{A}_{\text{BC}} = C^*(\mathbb{Q}/\mathbb{Z}) \rtimes \mathbb{N}^*,$$

built on $(\mathbb{N}^*, +, \times)$. Counting gives the objects, and countable objects admit only two ways of combining: merging gives addition, $|A \sqcup B| = |A| + |B|$, and independent pairing gives multiplication, $|A \times B| = |A| \cdot |B|$. The three experimental properties coincide with these one by one, and any system whose excitations are countable, whose energies add and whose independent composition multiplies can be described by the BC system.

This paper starts from its local decomposition: prime-by-prime reduction yields finite-dimensional state spaces, whose geometric properties are examined and on which a physical description is then established. We now turn to the definition of the BC system.

§1.1 Definition of the System

The BC system is completely determined by three items of arithmetic data:

- (i) the object set $\mathbb{N}^* = \{1, 2, 3, \dots\}$;
- (ii) unique factorization: $n = \prod_p p^{a_p(n)}$ (the fundamental theorem of arithmetic);
- (iii) the Hamiltonian: $H(n) = \log n$ – the nonnegative, regular, completely additive function on $\mathbb{N}^*$, unique up to positive scaling.

The following five structures are all determined by (i)–(iii).

- (iv) C^* -algebra: $\mathcal{A}_{\text{BC}} = C^*(e(r), \mu_n)$, with generators satisfying

$$\begin{aligned} e(r_1)e(r_2) &= e(r_1 + r_2), & \mu_n\mu_m &= \mu_{nm}, & \mu_n^*\mu_n &= 1, \\ e(r)\mu_n &= \mu_n e(nr) \text{ (covariance relation),} & \mu_n e(r)\mu_n^* &= \frac{1}{n} \sum_{ns=r} e(s) \text{ (transfer relation).} \end{aligned} \tag{1}$$

Uniquely determined by the arithmetic structure of (i)(ii).

- (v) Time evolution: $\sigma_t(\mu_n) = n^{it}\mu_n$, $\sigma_t(e(r)) = e(r)$ (uniquely determined by (iii): $n^{it} = e^{it \log n}$).

- (vi) Partition function:

$$Z(\beta) = \sum_n n^{-\beta} = \zeta(\beta) = \prod_p (1 - p^{-\beta})^{-1}, \quad \beta > 1. \tag{2}$$

Obtained by direct summation from (iii); the Euler product[19] is given by (ii).

(vii) KMS states[16, 17, 18]:

$$\phi_\beta(|n\rangle\langle m|) = \delta_{nm} \frac{n^{-\beta}}{\zeta(\beta)}, \quad \beta > 1. \quad (3)$$

The main theorem of Bost–Connes gives the complete classification of KMS states: for $\beta \leq 1$ the KMS state is unique; for $\beta > 1$ the extremal KMS states are parametrized by $\hat{\mathbb{Z}}^* \cong \text{Gal}(\mathbb{Q}^{\text{ab}}/\mathbb{Q})$ –symmetry breaking. Equation (3) is the Gibbs diagonal value shared by the extremal KMS states with $\beta > 1$ on matrix units.

(viii) Tensor product decomposition:

$$\ell^2(\mathbb{N}^*) \cong \widehat{\bigotimes_p \ell^2(\mathbb{Z}_{\geq 0})}, \quad |n\rangle \mapsto \bigotimes_p |a_p\rangle_p. \quad (4)$$

Uniquely determined by (ii); under this decomposition σ_t factorizes over the prime factors.

Uniqueness of $H = \log n$. Let $f : \mathbb{N}^* \rightarrow \mathbb{R}$ satisfy (a) $f(mn) = f(m) + f(n)$, (b) $f(n) \geq 0$, (c) $f(n+1) - f(n) \rightarrow 0$. Then $f = c \log n$, $c \geq 0$.

The argument proceeds in three steps.

Step 1. By (a), f is completely determined by its values on the primes: $f(n) = \sum_p a_p(n) f(p)$. (a)(b) alone do not suffice to fix f —each $f(p) \geq 0$ can be chosen independently.

Step 2. Regularity (c) is the key: the Erdős theorem (1946)[20] asserts that a completely additive function satisfying $f(n+1) - f(n) \rightarrow 0$ must be $f = c \log n$. The core mechanism: (c) forces the values at adjacent integers to agree asymptotically, while the prime-factor structures of adjacent integers are mutually independent, so the ratios $f(p)/\log p$ in all prime directions are locked to the same constant c .

Step 3. Condition (b) gives $c \geq 0$; $H = \log n$ corresponds to the scale $c = 1$ (the scale is absorbed into the time unit). $\square$

The above completely defines the BC system. The tensor product decomposition (viii) shows that the full system decomposes over primes into a product of local factors. The next section uses this decomposition to extract from the global system the finite-dimensional structure at each prime p —the full group $G_p^{(K)} = \mathbb{Z}/p^K\mathbb{Z}$.

§2 Local Decomposition and the Full Group

The tensor product decomposition (4) shows that the BC system can be analyzed prime by prime. The mathematical basis of this decomposition is Ostrowski’s theorem (1916)[21]: the nontrivial absolute values on $\mathbb{Q}$ fall into exactly two classes:

- (i) the Archimedean absolute value $|\cdot|_\infty$, giving the completion $\mathbb{R}$;
- (ii) for each prime p , the p -adic absolute value $|\cdot|_p$, giving the completion $\mathbb{Q}_p$.

Under this classification, $H(n) = \log n$ is precisely the logarithm of the Archimedean absolute value; the p -adic directions carry the local structure of the BC system at each prime p . This section extracts these local structures and truncates them to the finite-dimensional full group $G_p^{(K)} = \mathbb{Z}/p^K\mathbb{Z}$.

§2.1 Local Decomposition

The phase operators $e(r)$ of the BC algebra act on $\mathbb{Q}/\mathbb{Z}$. $\mathbb{Q}/\mathbb{Z}$ decomposes naturally over primes.

Direct sum decomposition of $\mathbb{Q}/\mathbb{Z}$.

$$\mathbb{Q}/\mathbb{Z} = \bigoplus_p \mathbb{Z}[1/p]/\mathbb{Z}, \quad (5)$$

where $\mathbb{Z}[1/p] = \{a/p^k : a \in \mathbb{Z}, k \geq 0\}$ is the localization of $\mathbb{Z}$ at p , and the direct sum runs over all primes p .

The argument proceeds in three steps.

Step 1. Any $r \in \mathbb{Q}/\mathbb{Z}$ can be written as $r = a/n$ ($a \in \mathbb{Z}, n \geq 1, 0 \leq a < n$). Let $n = p_1^{k_1} \cdots p_s^{k_s}$ be the unique factorization. By the Chinese remainder theorem[22],

$$\frac{a}{n} = \frac{a}{p_1^{k_1} \cdots p_s^{k_s}} \equiv \frac{a_1}{p_1^{k_1}} + \cdots + \frac{a_s}{p_s^{k_s}} \pmod{\mathbb{Z}},$$

where $a_i \in \{0, 1, \dots, p_i^{k_i} - 1\}$ is uniquely determined by CRT (using the modular inverse of $\prod_{j \neq i} p_j^{k_j}$). Hence r is uniquely written in $\mathbb{Q}/\mathbb{Z}$ as a sum of p -components $r = \sum_{i=1}^s r_{p_i}$, $r_{p_i} \in \mathbb{Z}[1/p_i]/\mathbb{Z}$.

Step 2. If $\sum_p s_p = 0$ ($s_p \in \mathbb{Z}[1/p]/\mathbb{Z}$, finitely many nonzero), then after clearing denominators the p -parts of the numerator are all 0, so each $s_p = 0$. Each $\mathbb{Z}[1/p]/\mathbb{Z}$ is a subgroup (closed under addition), and $\mathbb{Q}/\mathbb{Z} = \bigoplus_p \mathbb{Z}[1/p]/\mathbb{Z}$ holds as a direct sum of Abelian groups.

Step 3. $\mathbb{Z}[1/p]/\mathbb{Z} = \varinjlim (\mathbb{Z}/p^k\mathbb{Z})$, where the direct limit is given by $\mathbb{Z}/p^k\mathbb{Z} \hookrightarrow \mathbb{Z}/p^{k+1}\mathbb{Z}$ (the multiplication-by- p map). $\square$

Concrete construction of $\widehat{\mathbb{Z}}$. The elements of the projective limit $\widehat{\mathbb{Z}} = \varprojlim_n \mathbb{Z}/n\mathbb{Z}$ are compatible sequences:

$$\widehat{\mathbb{Z}} = \{(a_n)_{n \geq 1} \in \prod_n \mathbb{Z}/n\mathbb{Z} : m \mid n \Rightarrow a_n \equiv a_m \pmod{m}\}.$$

The argument proceeds in four steps.

Step 1. $\varprojlim \mathbb{Z}/n\mathbb{Z}$ is given by the limit cone of the directed system $\{\pi_{mn} : \mathbb{Z}/n\mathbb{Z} \rightarrow \mathbb{Z}/m\mathbb{Z}\}_{m|n}$ (the natural reduction maps). Concretely, consider the positive integers partially ordered by divisibility $m|n$; for each pair $m|n$ there is a homomorphism $\pi_{mn}(a \bmod n) = a \bmod m$. The elements of the projective limit are exactly the sequences satisfying the compatibility condition above.

Step 2. Componentwise, $(a_n) + (b_n) = (a_n + b_n)$; compatibility is verified directly: if $a_n \equiv a_m$ and $b_n \equiv b_m \pmod{m}$, then $(a_n + b_n) \equiv (a_m + b_m) \pmod{m}$.

Step 3. Similarly, $(a_n)(b_n) = (a_n b_n)$, yielding a ring.

Step 4. Equipped with the subspace topology of the product topology, $\widehat{\mathbb{Z}}$ is compact (Tychonoff's theorem). Typical elements: an integer $m \in \mathbb{Z}$ gives the constant sequence $(m \bmod n)_n$, so $\mathbb{Z} \hookrightarrow \widehat{\mathbb{Z}}$ is a dense embedding. $\square$

Product decomposition of the full profinite integer ring.

$$\widehat{\mathbb{Z}} = \varprojlim_n \mathbb{Z}/n\mathbb{Z} \cong \prod_p \mathbb{Z}_p, \quad (6)$$

where $\mathbb{Z}_p = \varprojlim_k \mathbb{Z}/p^k\mathbb{Z}$ is the ring of p -adic integers (the full ring, containing all p -adic integers, not just the units). The isomorphism holds both in the sense of rings and of topological groups.

The argument proceeds in three steps.

Step 1. For $n = \prod_i p_i^{k_i}$ (a finite product), the Chinese remainder theorem gives a ring isomorphism:

$$\mathbb{Z}/n\mathbb{Z} \xrightarrow{\sim} \prod_i \mathbb{Z}/p_i^{k_i}\mathbb{Z}, \quad a \bmod n \mapsto (a \bmod p_i^{k_i})_i.$$

The map is evidently a ring homomorphism; injectivity comes from CRT ($a \equiv 0 \pmod{p_i^{k_i}}$ for all i means $a \equiv 0 \pmod{n}$); surjectivity comes from $\gcd(p_i^{k_i}, p_j^{k_j}) = 1$ ($i \neq j$) and the solvability part of the Chinese remainder theorem.

Step 2. Taking projective limits over n (in the divisibility order) on both sides, the limit commutes with finite direct products:

$$\varprojlim_n \mathbb{Z}/n\mathbb{Z} \cong \varprojlim_n \prod_p \mathbb{Z}/p^{v_p(n)}\mathbb{Z} \cong \prod_p \varprojlim_k \mathbb{Z}/p^k\mathbb{Z} = \prod_p \mathbb{Z}_p.$$

Here the second step uses the interchange of (finite) direct products with projective limits (compatibility of categorical limits with products): for the primes in a fixed finite set S , $\varprojlim_n \prod_{p \in S} \mathbb{Z}/p^{v_p(n)}\mathbb{Z} = \prod_{p \in S} \mathbb{Z}_p$; then taking the topology (product topology) over all primes gives the full isomorphism.

Step 3. The map $\Phi : \widehat{\mathbb{Z}} \rightarrow \prod_p \mathbb{Z}_p$ is defined by: $(a_n)_{n \geq 1} \mapsto (a_{p^k})_{k \geq 1} \big|_{\text{each } p}$, i.e., for each prime p , extract the p -power components. This is a continuous ring isomorphism; the topological isomorphism follows because both sides are compact groups and the map is bijective and continuous. $\square$

Local decomposition of the BC system. Combining the Euler product $\zeta(\beta) = \prod_p (1 - p^{-\beta})^{-1}$ of §1 and the tensor product $\ell^2(\mathbb{N}^*) \cong \bigotimes_p \ell^2(\mathbb{Z}_{\geq 0})$, the BC system decomposes into local factors, one for each prime p :

$$\text{BC system} = \prod_p (\text{local system at } p).$$

Each local system corresponds to the structure of the full p -adic ring $\mathbb{Z}_p$.

§2.2 The Full Group

The local decomposition above reduces the BC system to independent factors at each prime p . Fix p ; the local system at p is built on $\mathbb{Z}_p$. Taking the K -digit truncation of $\mathbb{Z}_p$, define the **full group**

$$G_p^{(K)} = \mathbb{Z}/p^K\mathbb{Z} \quad (\text{as a ring, with } p^K \text{ elements}),$$

i.e., the natural truncation $\pi_K : \mathbb{Z}_p \twoheadrightarrow \mathbb{Z}/p^K\mathbb{Z}$. Note that what is taken here is the full ring $\mathbb{Z}/p^K\mathbb{Z}$, not the unit group $(\mathbb{Z}/p^K\mathbb{Z})^*$. The reason is as follows.

The phase operator $e(a/p^K)$ of the BC algebra is defined for all $\alpha \in \mathbb{Z}/p^K\mathbb{Z}$, acting as:

$$e(a/p^K) |\alpha\rangle = e^{2\pi i a \alpha / p^K} |\alpha\rangle, \quad \alpha \in \mathbb{Z}/p^K\mathbb{Z}.$$

This is equally well defined for $\alpha = 0, p, 2p, \dots, p^K - p$ (i.e., the multiples of p), giving special phases:

$$e(a/p^K) |mp\rangle = e^{2\pi i a m / p^{K-1}} |mp\rangle, \quad m = 0, 1, \dots, p^{K-1} - 1.$$

If only the unit group $\alpha \in (\mathbb{Z}/p^K\mathbb{Z})^*$ were taken (excluding the multiples of p), the crossed relations would break. A complete argument follows.

Step 1. The covariance relation $e(r) \mu_n = \mu_n e(nr)$ requires: for each $r = a/p^K$ and each $n \in \mathbb{N}^*$, μ_n maps $|\alpha\rangle$ (α a unit) to some $|\alpha'\rangle$ (α' should also lie in the state space).

Step 2. Under the natural action on $\mathbb{Z}/p^K\mathbb{Z}$, μ_p maps α to $p\alpha \bmod p^K$. Take $\alpha = 1$: $\mu_p |1\rangle = |p\rangle$. But $p \cdot 1 = p \notin (\mathbb{Z}/p^K\mathbb{Z})^*$ (p is not a unit).

Step 3. If the state space contained only $\{|\alpha\rangle : \gcd(\alpha, 3) = 1\}$, take $n = 3$, $r = 1/3^5$. Acting with both sides of the covariance relation on $|1\rangle$:

$$e(1/3^5) \mu_3 |1\rangle = e(1/3^5) |3\rangle, \quad \mu_3 e(3/3^5) |1\rangle = e^{2\pi i / 3^4} \mu_3 |1\rangle = e^{2\pi i / 3^4} |3\rangle,$$

both sides require $\mu_3 |1\rangle = |3\rangle$ to exist in the state space— but $3 \equiv 0 \pmod{3}$, i.e., 3 is a multiple of 3 in $\mathbb{Z}/3^K\mathbb{Z}$ and does not belong to the unit group. If $|3\rangle$ is not in the state space, the covariance relation $e(r) \mu_3 = \mu_3 e(3r)$ cannot hold at $|1\rangle$ (the resulting vector lies outside the state space), signaling a break.

Hence the complete representation of the BC algebra requires the state space to contain the multiples of p , and the full group must be the full ring $\mathbb{Z}/p^K\mathbb{Z}$ rather than the unit group $(\mathbb{Z}/p^K\mathbb{Z})^*$. $\square$

§2.3 The Choice of K

The full-ring truncation $\mathbb{Z}/p^k\mathbb{Z}$ is defined for every $k \geq 1$. The derivation of the five axioms in this paper is valid for *any fixed* K ; for definiteness, we take

$$K \equiv K_{\max} = \dim_{\mathbb{F}_p} I = p(p-1) - 1, \quad (7)$$

where I is the augmentation ideal of the group algebra of the affine group $\text{Aff}(\mathbb{F}_p) = \mathbb{F}_p \rtimes \mathbb{F}_p^*$ ($|\text{Aff}(\mathbb{F}_p)| = p(p-1)$): $K_{\max}$ is the natural truncation at which the full ring completely accommodates the group of combined additive and multiplicative transformations of $\mathbb{F}_p$ (affine completeness).

§2.4 The Unit Group and the Teichmüller Decomposition

The p^K elements of the full ring fall, according to p -adic valuation, into the units $(\mathbb{Z}/p^K\mathbb{Z})^*$ (coprime to p , $\varphi(p^K)$ in total) and the multiples of p . For odd primes p , the unit group admits the canonical Teichmüller splitting

$$(\mathbb{Z}/p^K\mathbb{Z})^* \cong \mu_{p-1} \times (1 + p\mathbb{Z}/p^K\mathbb{Z}), \quad \mu_{p-1} \cong (\mathbb{Z}/p\mathbb{Z})^* \cong \mathbb{Z}/(p-1)\mathbb{Z}. \quad (8)$$

The Teichmüller part μ_{p-1} is the domain of the multiplicative characters of §3, whose CRT decomposition gives the tensor-product structure of Axiom III.

§3 Algebraic Structure of the State Space

§2.2 determined that the underlying set of the BC algebra at the prime p is the full ring $\mathbb{Z}/p^K\mathbb{Z}$ (the crossed relations require the multiples of p to be included), and §2.3 determined $K_{\max} = p(p-1) - 1$ (the state space takes $K \equiv K_{\max}$). To build a state space on the full ring, an inner-product structure is needed to define orthogonality. The function space on the full ring, equipped with the standard inner product, gives the **state space**:

$$\boxed{\mathcal{H}_p = L^2(\mathbb{Z}/p^K\mathbb{Z}, \mathbb{C}), \quad \dim \mathcal{H}_p = p^K.} \quad (9)$$

Inner product: for $f, g \in \mathcal{H}_p$,

$$\langle f, g \rangle = \frac{1}{p^K} \sum_{\alpha \in \mathbb{Z}/p^K\mathbb{Z}} \overline{f(\alpha)} g(\alpha).$$

$\mathcal{H}_p$ is a p^K -dimensional Hilbert space. Finite dimensionality guarantees that all operators are bounded and all spectra are completely discrete—the entire spectral structure of the state space can be described by finite matrices.

The additive Fourier basis

$$\chi_j^{(\text{add})}(\alpha) = e^{2\pi i j \alpha / p^K}, \quad j = 0, 1, \dots, p^K - 1,$$

forms an orthonormal basis of $\mathcal{H}_p$: $\langle \chi_j^{(\text{add})}, \chi_k^{(\text{add})} \rangle = \delta_{jk}$. Besides the additive Fourier basis, the state space carries a second family of basis functions—the multiplicative characters, defined on the unit group $(\mathbb{Z}/p\mathbb{Z})^*$.

This section establishes the algebraic structure of the state space: the multiplicative characters and their properties, the spectral structure, symmetry, and metric.

§3.1 Multiplicative Characters

A **multiplicative character** [23] of $(\mathbb{Z}/p\mathbb{Z})^*$ is a group homomorphism $\chi : (\mathbb{Z}/p\mathbb{Z})^* \rightarrow \mathbb{C}^*$, $\chi(ab) = \chi(a)\chi(b)$. Since $(\mathbb{Z}/p\mathbb{Z})^*$ is a cyclic group of order $p-1$ (with generator g), all multiplicative characters are given by $\chi_k(g) = e^{2\pi i k/(p-1)}$ ($k = 0, \dots, p-2$), $p-1$ in total. The subsequent subsections describe the properties of these $p-1$ characters in turn: order and conjugation, and CRT coordinates.

The multiplicative character χ_k is embedded into $\mathcal{H}_p$ as follows:

$$\tilde{\chi}_k(\alpha) = \begin{cases} \chi_k(\alpha \bmod p), & \text{if } \gcd(\alpha, p) = 1, \\ 0, & \text{if } p \mid \alpha. \end{cases} \quad (10)$$

That is, the character value is taken on the units, and 0 on the multiples of p . Comparison of the two kinds of characters:

	Additive character	Multiplicative character
Domain	$\mathbb{Z}/p^K\mathbb{Z}$ (full group)	$(\mathbb{Z}/p\mathbb{Z})^*$ (unit group)
Homomorphism property	$\chi(a+b) = \chi(a)\chi(b)$	$\chi(ab) = \chi(a)\chi(b)$
Number	p^K	$p-1$

The additive characters form a complete orthogonal basis of L^2 [24]; the multiplicative characters are embedded into L^2 via (10) and carry the multiplicative structure of the full group.

3.1.1 Order and Conjugation

The **order** of a multiplicative character χ_k is the smallest positive integer d with $\chi_k^d = \chi_0$, namely $d = (p-1)/\gcd(k, p-1)$; for each $d \mid (p-1)$ there are exactly $\varphi(d)$ characters of order d (φ the Euler function, $\sum_{d \mid p-1} \varphi(d) = p-1$). Conjugation: $\overline{\chi_k} = \chi_{-k}$; χ_k is a real character if and only if its order is ≤ 2 .

3.1.2 CRT Decomposition

Let $p-1 = q_1^{b_1} \cdots q_s^{b_s}$. Then the character group decomposes as

$$\mathbb{Z}/(p-1)\mathbb{Z} \cong \mathbb{Z}/q_1^{b_1}\mathbb{Z} \times \cdots \times \mathbb{Z}/q_s^{b_s}\mathbb{Z}, \quad (11)$$

inducing a tensor product of the mode space: $L^2(\mu_{p-1}) \cong \bigotimes_{i=1}^s L^2(\mathbb{Z}/q_i^{b_i}\mathbb{Z})$.

The argument proceeds in three steps.

Step 1. The q_i are pairwise coprime; the Chinese remainder theorem gives a group isomorphism: $\mathbb{Z}/(p-1)\mathbb{Z} \cong \mathbb{Z}/q_1^{b_1}\mathbb{Z} \times \cdots \times \mathbb{Z}/q_s^{b_s}\mathbb{Z}$, with map $a \mapsto (a \bmod q_1^{b_1}, \dots, a \bmod q_s^{b_s})$.

Step 2. For a product group $G = \prod_i G_i$ ($G_i = \mathbb{Z}/q_i^{b_i}\mathbb{Z}$), Pontryagin duality[25] gives $\widehat{G} \cong \prod_i \widehat{G_i}$. Each character decomposes uniquely as $\chi = \chi_1 \otimes \cdots \otimes \chi_s$.

Step 3. The orthogonal basis $\{\chi\}_{\chi \in \widehat{G}}$ of $L^2(G)$ corresponds bijectively via CRT to the basis $\{\chi_1 \otimes \cdots \otimes \chi_s\}$ of $\bigotimes_i L^2(G_i)$. The inner product is preserved:

$$\langle \chi_1 \otimes \cdots \otimes \chi_s, \chi'_1 \otimes \cdots \otimes \chi'_s \rangle = \prod_i \delta_{\chi_i, \chi'_i}.$$

Hence $L^2(G) \cong \bigotimes_i L^2(G_i)$ is an isomorphism of Hilbert spaces. $\square$

The mode space thus decomposes into $s = \omega(p-1)$ independent directions.

§3.2 Spectral Structure

3.2.1 The Convolution Operator

For $f \in \mathcal{H}_p$, the **convolution operator** K is defined as the KMS-weighted sum of scaling operators:

$$(Kf)(\alpha) = \sum_{\substack{n \geq 1 \\ p \nmid n}} n^{-\beta} f(n\alpha), \quad \alpha \in \mathbb{Z}/p^K\mathbb{Z}, \quad \beta > 1. \quad (12)$$

$\beta > 1$ guarantees absolute convergence of the series, so K is bounded; $p \nmid n$ guarantees that $\alpha \mapsto n\alpha$ is invertible; the weights $n^{-\beta}$ come from the KMS state, and the scaling action comes from the covariance relation $e(r)\mu_n = \mu_n e(nr)$. K is a bounded linear operator on $\mathcal{H}_p$.

3.2.2 Diagonalization Theorem

The convolution operator K is diagonalized in the multiplicative character basis $\{\tilde{\chi}_k\}_{k=0}^{p-2}$ ((10)):

$$K\tilde{\chi}_k = \hat{K}(\chi_k) \cdot \tilde{\chi}_k, \quad (13)$$

with eigenvalues the Dirichlet L -functions[27]:

$$\boxed{\hat{K}(\chi_k) = L(\beta, \chi_k) = \sum_{\substack{n \geq 1 \\ p \nmid n}} \chi_k(n) n^{-\beta}.} \quad (14)$$

In particular, $\hat{K}(\chi_0) = \zeta(\beta) \cdot (1 - p^{-\beta})$ (the ζ function with the Euler factor at p removed).

Argument. $\tilde{\chi}_k(\alpha) = \chi_k(\alpha \bmod p)$ takes the character value on units and vanishes on non-units. For $p \nmid n$, α and $n\alpha$ are both units or both non-units, and $\tilde{\chi}_k(n\alpha) = \chi_k(n) \tilde{\chi}_k(\alpha)$. Substituting into (12):

$$(K\tilde{\chi}_k)(\alpha) = \sum_{p \nmid n} n^{-\beta} \chi_k(n) \tilde{\chi}_k(\alpha) = L(\beta, \chi_k) \tilde{\chi}_k(\alpha). \quad \square$$

3.2.3 Structure of the Eigenvalue Spectrum

For each prime p , K has $p - 1$ eigenvalues in the multiplicative directions, $\{\lambda_k = L(\beta, \chi_k) : k = 0, \dots, p - 2\}$:

- (i) $\lambda_0 = L(\beta, \chi_0) \propto \zeta(\beta)$ (trivial character, proportional to the partition function);
- (ii) for real characters ($\chi_k = \bar{\chi}_k$): $\lambda_k \in \mathbb{R}$;
- (iii) for complex characters ($\chi_k \neq \bar{\chi}_k$): $\lambda_k \in \mathbb{C}$, with $\lambda_{\bar{k}} = \overline{\lambda_k}$ (the eigenvalues of conjugate characters are complex conjugates of each other).

§3.3 Symmetry: the Galois Action

The natural symmetry of the state space $\mathcal{H}_p$ relevant to the five axioms is the Galois action σ .

The multiplicative characters take values in the $(p - 1)$ -th roots of unity, lying in the cyclotomic field $\mathbb{Q}(\zeta_{p-1})$ [26]. The Galois group acting on the character values is $\text{Gal}(\mathbb{Q}(\zeta_{p-1})/\mathbb{Q}) \cong (\mathbb{Z}/(p - 1)\mathbb{Z})^*$, acting by

$$\sigma_a(\chi_k) = \chi_k^a = \chi_{ak \bmod (p-1)} \quad (a \in (\mathbb{Z}/(p - 1)\mathbb{Z})^*).$$

$\gcd(a, p - 1) = 1$ guarantees that $k \mapsto ak$ is a bijection of $\mathbb{Z}/(p - 1)\mathbb{Z}$ preserving the order of each element, so this is a group action; a full Galois orbit consists of characters of the same order, of size $\varphi(d)$ (d the order of χ_k).

Key special case: -1 is always coprime to $p - 1$, and $\sigma_{-1}(\chi_k) = \chi_{-k} = \bar{\chi}_k$ is complex conjugation. The orbits of the subgroup $\{1, \sigma_{-1}\}$ are the **conjugate orbits**, of size

$$G(\chi_k) = \begin{cases} 2, & \chi_k \neq \bar{\chi}_k \text{ (complex character, orbit } \{\chi_k, \bar{\chi}_k\}), \\ 1, & \chi_k = \bar{\chi}_k \text{ (real character, fixed point);} \end{cases}$$

equivalently $G(\chi_k) = [\mathbb{Q}(\chi_k) : \mathbb{Q}(\chi_k) \cap \mathbb{R}]$, the degree of the character value field over its maximal real subfield. It is on these orbits that §4.2 takes invariants.

§3.4 Metric Structure

The eigenvalues of the Hamiltonian $H = \log n$ embed the state space into the real line, $n \mapsto \log n$. The same eigenvalues carry three scales: the distance $d(m, n) = |\log m - \log n|$, the energy $E_n = \log n$, and, through $\sigma_t = e^{itH}$, the period $T_n = 2\pi/\log n$ ($n \geq 2$). The three scales share the single spacing $\log(1 + 1/n)$.

That $(\mathbb{N}^*, d)$ is a metric space is verified item by item. Positive definiteness: $\log$ is strictly increasing on $\mathbb{N}^*$, so $m \neq n$ gives $|\log m - \log n| > 0$, and $d(n, n) = 0$. Symmetry: $|\log m - \log n| = |\log n - \log m|$. Triangle inequality: inserting $\log k$ and applying the triangle inequality for the absolute value on $\mathbb{R}$,

$$|\log m - \log n| = |(\log m - \log k) + (\log k - \log n)| \leq d(m, k) + d(k, n).$$

□

The adjacent spacing $d(n, n+1) = \log(1 + 1/n)$ attains its maximum $\log 2$ at $n = 1$ and decreases strictly monotonically to zero. The state space is thus coarse at the bottom and asymptotically fine at the top: quantum effects are prominent in the low-energy region, while the high-energy region approaches the classical continuum limit.

The energy spectrum $\{\log n\}$ is discrete, with minimal excitation energy $\log 2$. Energy is additive with respect to multiplication, $E_{mn} = E_m + E_n$, and prime factorization decomposes an excitation into its indivisible constituents. These properties depend on $H = \log n$ alone, whose uniqueness is guaranteed by the Erdős theorem of §1; the additivity will be used again in §4.4.

§4 Quantum Structure of the State Space

§3 established the complete algebraic structure of the state space $\mathcal{H}_p$: the multiplicative characters as the mode basis, the eigenvalues of the convolution operator as L -functions, symmetry given by the Galois action, and the metric given by $H = \log n$. Taking these structures as given, this section verifies the five axioms of von Neumann quantum mechanics[29] one at a time.

§4.1 Hilbert Space

von Neumann (1932) requires the pure states of a physical system to form rays in a Hilbert space $\mathcal{H}$, guaranteeing the superposition principle, the probability interpretation, and spectral decomposition. For the BC system, $\mathcal{H}_p$ is constructed as follows.

The argument proceeds in four steps.

Step 1. The C^* -algebra axioms give $\phi(a^*a) \geq 0$ for all $a \in \mathcal{A}_{\text{BC}}$. Define $\langle a, b \rangle_\phi = \phi(a^*b)$. This map automatically satisfies:

- conjugate symmetry: $\langle a, b \rangle = \overline{\langle b, a \rangle}$ (by the $*$ -preservation of ϕ);
- linearity: $\langle a, \alpha b + \beta c \rangle = \alpha \langle a, b \rangle + \beta \langle a, c \rangle$ (by the linearity of ϕ);
- positive definiteness: $\langle a, a \rangle = \phi(a^*a) \geq 0$, with equality if and only if $a \in \mathcal{N} = \{a : \phi(a^*a) = 0\}$.

Step 2. Take the quotient $\mathcal{A}/\mathcal{N}$ by the null space $\mathcal{N}$; on it $\langle \cdot, \cdot \rangle$ becomes a strictly positive-definite inner product. Completing with respect to this inner product gives the Hilbert space $\mathcal{H}_\phi$.

Step 3. For the BC system, for $\beta > 1$ the KMS state gives $\phi_\beta(|n\rangle\langle n|) = n^{-\beta}/\zeta(\beta) > 0$ for all $n \in \mathbb{N}^*$, so $\mathcal{N} = \{0\}$ (no null vector is quotiented away). The GNS space $\mathcal{H}_\phi$ [11, 12, 15] is of Hilbert–Schmidt type (basis $\{e_{nm}\Omega\}$; the modular structure in Step 6 of §4.5 acts precisely on this space), and its diagonal slice gives the original representation space $\ell^2(\mathbb{N}^*)$. After p -adic reduction (§2) one obtains the finite-dimensional local space $\mathcal{H}_p = L^2(\mathbb{Z}/p^K\mathbb{Z})$.

Step 4. $\dim \mathcal{H}_p = p^K < \infty$. A finite-dimensional normed space is automatically complete (no separate verification of the convergence of Cauchy sequences is needed), so $\mathcal{H}_p$ is a Hilbert space. $\square$

§4.2 Self-Adjoint Observables

With the Hilbert space established, the next question is the structure of observables. von Neumann requires observables to be represented by self-adjoint operators $A = A^*$, guaranteeing real measurement results. In the BC framework, observables are selected by conjugation invariance.

Let O be an operator on $\mathcal{H}_p$ with the χ_k as eigenvectors and eigenvalues λ_k . A physical observable is taken as $O^\dagger O$, whose eigenvalues $|\lambda_k|^2 \in \mathbb{R}_{\geq 0}$ are real and non-negative; $O^\dagger O$ is self-adjoint.

The argument proceeds in four steps.

Step 1. The Galois group $\text{Gal}(\mathbb{Q}(\zeta_{p-1})/\mathbb{Q}) \cong (\mathbb{Z}/(p-1)\mathbb{Z})^*$ of §3.3 acts on the character labels: $\sigma_a(\chi_k) = \chi_{ak \bmod (p-1)}$. If an operator O satisfies $O\chi_k = \lambda_k\chi_k$, this action induces a label permutation of the eigenvalues $\lambda_k \mapsto \lambda_{ak}$ (a rearrangement of the spectrum). Note that σ_a permutes the labels; it does not move λ_k around as an algebraic number—the L -function values themselves do not belong to any cyclotomic field.

Step 2. Physically observable values must be invariant under the label action of the conjugation subgroup $\{1, \sigma_{-1}\}$ (§3.3): σ_{-1} exchanges $\chi_k \leftrightarrow \bar{\chi}_k$, corresponding to $\lambda_k \mapsto \lambda_{-k} = \bar{\lambda}_k$ (the complex-conjugation relation of L -functions). For complex characters ($\chi_k \neq \bar{\chi}_k$), $\lambda_k \neq \bar{\lambda}_k$, so λ_k itself is not conjugation-invariant—it cannot serve directly as an observable.

Step 3. The norm of λ_k with respect to $\{1, \sigma_{-1}\}$ is $N(\lambda_k) = \lambda_k \cdot \sigma_{-1}(\lambda_k) = \lambda_k \cdot \bar{\lambda}_k = |\lambda_k|^2 \in \mathbb{R}$ —invariant under the conjugation action, and an eigenvalue of $O^\dagger O$.

Step 4. The observable is $O^\dagger O$, with eigenvalues $|\lambda_k|^2 \in \mathbb{R}_{\geq 0}$; it is self-adjoint. $\square$

§4.3 Tensor Product

The state space of a composite system is required to have a tensor product structure $\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B$.

For finite groups G and H , the map $\delta_{(g,h)} \mapsto \delta_g \otimes \delta_h$ carries an orthonormal basis of $L^2(G \times H)$ to an orthonormal basis of $L^2(G) \otimes L^2(H)$, so that

$$L^2(G \times H) \cong L^2(G) \otimes L^2(H), \quad \dim = |G| \cdot |H|. \quad (15)$$

When G and H are abelian, Pontryagin duality gives $\widehat{G \times H} \cong \widehat{G} \times \widehat{H}$, the characters multiplying pointwise.

The additive group $\mathbb{Z}/p^K\mathbb{Z}$ of $\mathcal{H}_p$ is cyclic of prime-power order, admits no co-prime factors, and is a single degree of freedom. The tensor product structure appears in the multiplicative direction: by (8) the unit group splits canonically as $\mu_{p-1} \times (1 + p\mathbb{Z}/p^K\mathbb{Z})$, where μ_{p-1} is the domain of the multiplicative characters.

Let $p-1 = q_1^{b_1} \cdots q_s^{b_s}$. Since μ_{p-1} is cyclic and a cyclic group has exactly one subgroup of each order, the decomposition into coprime subgroups $\mu_{p-1} \cong \prod_i \mathbb{Z}/q_i^{b_i}\mathbb{Z}$ exists and is unique, whence

$$L^2(\mu_{p-1}) \cong \bigotimes_{i=1}^s L^2(\mathbb{Z}/q_i^{b_i}\mathbb{Z}). \quad (16)$$

□

§4.4 Unitary Time Evolution

With the state space, observables, and composite structure determined, time evolution is still needed. Time evolution must preserve probability, i.e., form a unitary group. The unitary evolution σ_t ($t \in \mathbb{R}$) of the BC system is defined by:

$$(\sigma_t f)(\alpha) = e^{it \cdot v_p(\alpha)} f(\alpha),$$

where v_p is the p -adic valuation. σ_t preserves the inner product, satisfies the group law, and has H as its generator.

The argument proceeds in four steps.

Step 1. (Unitarity)

$$\begin{aligned} \langle \sigma_t f, \sigma_t g \rangle &= \frac{1}{p^K} \sum_{\alpha \in \mathbb{Z}/p^K\mathbb{Z}} \overline{e^{it \cdot v_p(\alpha)} f(\alpha)} \cdot e^{it \cdot v_p(\alpha)} g(\alpha) \\ &= \frac{1}{p^K} \sum_{\alpha} \overline{f(\alpha)} g(\alpha) = \langle f, g \rangle. \end{aligned}$$

Key point: $|e^{it \cdot v_p(\alpha)}| = 1$ ($v_p(\alpha) \in \mathbb{Z}_{\geq 0}$, a pure phase), and the phase factors cancel.

Step 2. (Group law)

$$(\sigma_s \circ \sigma_t f)(\alpha) = e^{is \cdot v_p(\alpha)} \cdot e^{it \cdot v_p(\alpha)} f(\alpha) = e^{i(s+t) \cdot v_p(\alpha)} f(\alpha) = (\sigma_{s+t} f)(\alpha).$$

Together with $\sigma_0 = \text{id}$, $\{\sigma_t\}_{t \in \mathbb{R}}$ forms a one-parameter unitary group[13].

Step 3. (Generator) $\sigma_t = e^{itH}$, $H(\alpha) = v_p(\alpha)$. H is self-adjoint (v_p is real-valued), so σ_t is unitary.

Step 4. (Uniqueness) Let σ'_t be another one-parameter group of $*$ -automorphisms of $\mathcal{A}_{\text{BC}}$. Restricted to the phase subalgebra $C^*(\mathbb{Q}/\mathbb{Z})$, σ'_t gives a continuous homomorphism $\mathbb{R} \rightarrow \hat{\mathbb{Z}}^*$; $\hat{\mathbb{Z}}^*$ is totally disconnected, so the continuous image must be trivial, and σ'_t fixes each $e(r)$ pointwise. The BC algebra relations then force $\sigma'_t(\mu_n) = \lambda_n(t)\mu_n$; complete multiplicativity gives $\lambda_{mn}(t) = \lambda_m(t)\lambda_n(t)$, unitarity gives $|\lambda_n(t)| = 1$, and the group law gives the Cauchy functional equation $\lambda_n(s+t) = \lambda_n(s)\lambda_n(t)$, so $\lambda_n(t) = n^{ict}$; the KMS condition unifies the phases at all primes to a single scale: $\lambda_n(t) = n^{ict}$. The scale c is absorbed into the time unit; take $c = 1$ (consistent with the scale convention of §1(iii)). Hence σ_t is the unique time evolution. □

§4.5 The Born Rule

The standard statement of the Born (1926) rule[28]: a quantum system is in the state $|\psi\rangle$; measuring an observable A (eigenvalues $\{a_n\}$, eigenstates $\{|a_n\rangle\}$) yields the result a_n with probability

$$P(a_n) = |\langle a_n | \psi \rangle|^2 = \langle \psi | P_n | \psi \rangle,$$

where $P_n = |a_n\rangle\langle a_n|$ is the projection onto the eigendirection. For a mixed state ρ : $P(a_n) = \text{Tr}(\rho P_n)$.

For the BC system, the state is the KMS state φ_β , the observable directions are the characters χ_k , with corresponding character component operators P_{χ_k} (Step 1 below). The C^* positive functional φ_β provides nonnegativity ($\varphi_\beta(P^*P) \geq 0$) and additivity; normalization is completed by the explicit normalization in Step 5. The specific form of $\varphi_\beta(P_{\chi_k})$ is determined below.

The argument proceeds in six steps.

Step 1. Define the **component operator** (Fourier coefficient operator) of the character direction χ_k :

$$P_{\chi_k} = \frac{1}{p-1} \sum_{a=1}^{p-1} \overline{\chi_k(a)} e(a/p),$$

where $e(a/p)$ are the additive generators (phase operators) of the BC algebra. P_{χ_k} extracts the χ_k -Fourier component but is not an idempotent projection ($P^2 \neq P$)—its eigenvalues on the basis $|\alpha\rangle$ have modulus $\sqrt{p}/(p-1) \notin \{0, 1\}$.

Step 2. The Gibbs form $\phi_\beta = \text{Tr}(e^{-\beta H} \cdot) / \zeta(\beta)$ is evaluated directly on the phase operators: for $\text{gcd}(a, p) = 1$,

$$\phi_\beta(e(a/p)) = \frac{1}{\zeta(\beta)} \sum_{n \geq 1} n^{-\beta} e^{2\pi i n a / p} = \frac{1}{\zeta(\beta)} \sum_{\substack{n \geq 1 \\ p \nmid n}} n^{-\beta} e^{2\pi i n a / p} + p^{-\beta},$$

where the last term comes from the modes with $p \mid n$ (on which $e^{2\pi i n a / p} = 1$, the sum being $p^{-\beta} \zeta(\beta)$).

Step 3. Substitute Step 2 into the definition of P_{χ_k} . The contribution of the $p \mid n$ terms is weighted by $\sum_a \overline{\chi_k(a)}$: for nontrivial χ_k it is annihilated by character orthogonality $\sum_a \overline{\chi_k(a)} = 0$, leaving the constant term $p^{-\beta}$ only in the χ_0 direction. For the nontrivial directions:

$$\phi_\beta(P_{\chi_k}) = \frac{1}{(p-1)\zeta(\beta)} \sum_{\substack{n \geq 1 \\ p \nmid n}} n^{-\beta} \underbrace{\sum_{a=1}^{p-1} \overline{\chi_k(a)} e^{2\pi i n a / p}}_{S_k(n)}. \quad (17)$$

Step 4. For $\text{gcd}(n, p) = 1$, make the substitution $a = n^{-1}b \pmod{p}$ (n^{-1} exists since $\text{gcd}(n, p) = 1$):

$$S_k(n) = \sum_{b=1}^{p-1} \overline{\chi_k(n^{-1}b)} e^{2\pi i n b / p} = \chi_k(n) \sum_{b=1}^{p-1} \overline{\chi_k(b)} e^{2\pi i b / p} = \chi_k(n) \cdot \chi_k(-1) \overline{\tau(\chi_k)},$$

where $\tau(\chi_k) = \sum_{b=1}^{p-1} \chi_k(b) e^{2\pi i b/p}$ is the Gauss sum, $\tau(\bar{\chi}_k) = \chi_k(-1) \overline{\tau(\chi_k)}$. Substituting back:

$$\phi_\beta(P_{\chi_k}) = \frac{\chi_k(-1) \overline{\tau(\chi_k)}}{(p-1)\zeta(\beta)} \cdot L(\beta, \chi_k), \quad (18)$$

where the sign factor $\chi_k(-1) = \pm 1$ has no effect after taking the modulus squared. Here $L(\beta, \chi_k) = \sum_{n \geq 1, p \nmid n} \chi_k(n) n^{-\beta}$.

The Dirichlet L -function appears as the evaluation of the KMS state on the Fourier component.

Step 5. $\phi_\beta(P_{\chi_k}) \propto L(\beta, \chi_k)$ is the evaluation of the KMS state on the Fourier component; for complex characters it is a complex number, not a probability. The probability is given by the vertex vector of Step 6, as the modulus squared of the amplitude, invariant under the conjugation action $\lambda_k \mapsto \bar{\lambda}_k$ (§4.2). Define the response and the measurement probability:

$$R_k(\beta) = \frac{L(\beta, \chi_k)}{\zeta(\beta)}, \quad P(\chi_k) = \frac{|R_k(\beta)|^2}{\sum_j |R_j(\beta)|^2}.$$

The explicit normalization guarantees $\sum_k P(\chi_k) = 1$ for any β -normalization itself does not select the evaluation point. The evaluation point is determined by the modular structure of the state space; this is the content of Step 6.

Step 6 (evaluation point: modular structure). For $\beta > 1$, ϕ_β is a Gibbs state, its GNS representation is in standard form, and the Tomita–Takesaki theorem[14] gives the polar decomposition of the antilinear Tomita operator

$$S = J\Delta^{1/2}, \quad S a \Omega = a^* \Omega,$$

where J is the modular conjugation (antiunitary) and Δ the modular operator (positive), both uniquely determined by the uniqueness of the polar decomposition. The coefficients of the GNS vector $\Omega = \rho^{1/2}$ ($\rho = e^{-\beta H} / \zeta(\beta)$) are $n^{-\beta/2} / \sqrt{\zeta(\beta)}$: the amplitude-level counterpart of the probability-level weight $n^{-\beta}$ is $n^{-\beta/2}$, the weight at half the imaginary time $\beta/2$; $\Delta^{1/2}(a\Omega) = \rho^{1/2} a \rho^{-1/2} \Omega$ implements this half step, acting as the identity on the diagonal operators commuting with H and giving $n^{-\beta/2}$ on the transition units $|n\rangle\langle 1|$. The half exponent is not a choice; it is the unique positive part of the polar decomposition.

The evaluation has two levels: the complete pairing with the KMS state is at the probability level, and the pairing with the GNS vector Ω is at the amplitude level. A vertex is evaluated at the amplitude level: the direction is read out by the normalized character filter $\hat{P}_\chi = P_\chi / (\sqrt{p}/(p-1))$ (eigenvalues of modulus 1), and with $\mathbf{1}$ the flat vector giving equal weight to all modes, the vertex amplitude is

$$\langle \mathbf{1} | \hat{P}_\chi \Omega \rangle = e^{i\theta_\chi} \frac{1}{\sqrt{\zeta(\beta)}} \sum_n \chi(n) n^{-\beta/2},$$

and the product of the two halves (one factor Ω on each side of the pairing) returns to the probability level, $\langle \Omega | \hat{P}_\chi \Omega \rangle = e^{i\theta_\chi} L(\beta, \chi) / \zeta(\beta)$. The evaluation point is uniquely determined by three requirements: complete multiplicativity and measurability of the weight give $w(n) = n^{-s}$ for some s ; lying at the amplitude level (the $\Delta^{1/2}$ layer) forces $s = \beta/2$ (uniqueness of the polar decomposition); compatibility with the KMS structure at $\beta = 1$ takes $\beta = 1$ —hence $s = 1/2$ is unique.

The structure above holds pointwise for $\beta > 1$; $\beta = 1$ is the critical point of the phase transition ($\zeta(1)$ diverges), and the limit $\beta \rightarrow 1^+$ is examined character direction by character direction: the nontrivial χ components $\sum_n \chi(n) n^{-\beta/2} \rightarrow L(1/2, \chi)$ (the Dirichlet series of non-principal characters converges for $\text{Re } s > 0$), so the limit is well defined; the amplitude of the principal direction is $\sum_{n \leq N} n^{-1/2} = 2\sqrt{N} + \zeta(1/2) + O(N^{-1/2})$, the part $2\sqrt{N}$ growing with the truncation depth being extensive, and the depth-independent part being $\zeta(1/2)$. The amplitude-level divergence of the principal direction is the signal of collective condensation: elementary vertex responses exist only in the nontrivial directions; the χ_0 direction has no elementary vertex, only collective emergence; the vertex is normalized by the depth-independent part $\zeta(1/2)$ of the background amplitude.

$1/2$ is not another temperature—there is no trace-class Gibbs density at $\beta = 1/2$. It is the midpoint of the KMS analytic strip $[0, i\beta]$ of the $\beta = 1$ state, i.e., the location where the modular conjugation J acts; probability is the modulus squared of amplitude, and probability level $\beta = 1$ corresponds to amplitude level $\beta = 1/2$.

Final result:

$$R_k = \frac{L(1/2, \chi_k)}{\zeta(1/2)}, \quad P(\chi_k) = \frac{|R_k|^2}{\sum_j |R_j|^2}. \quad (19)$$

$P(\chi_k)$ is the modulus squared of a component of a pure vector, and the explicit normalization guarantees $\sum_k P(\chi_k) = 1$. In standard quantum mechanics the Born rule is regarded as an independent postulate—about the probabilities of measurement outcomes. The construction above shows: probability is the modulus squared of the component of the amplitude-level pairing along a character direction; the half weights and the L -value amplitudes are given by the modular structure, while the vertex being evaluated at the amplitude level, the direction being read out by characters, and the principal direction being normalized by the depth-independent part are three structural conditions. Probability is an objective relation between two states— not the output of a measurement, not the belief of an observer, not an intrinsic property of a particle.

§5 Conclusion

At this point all five axioms have been given by the structure of the state space; the correspondence is as follows.

Axiom	Content	State-space structure
I	Hilbert space	C^* -positivity $\rightarrow$ GNS $\rightarrow L^2(\mathbb{Z}/p^K\mathbb{Z})$ (§4.1)
II	Self-adjoint observables	Galois conjugation-invariant combination $ \lambda_k ^2$ (§4.2)
III	Tensor product	μ_{p-1} cyclic + CRT $\rightarrow$ decomposition unique (§4.3)
IV	Unitary evolution	complete multiplicativity + Cauchy + ζ monotonicity $\rightarrow \sigma_t$ unique (§4.4)
V	Born rule	two evaluation points of $S = J\Delta^{1/2} \rightarrow P = R_k ^2 / \sum R_j ^2$ (§4.5)

§5.1 Closing Remarks

Starting from the Bost–Connes system, the p -adic reduction gives the state space $\mathcal{H}_p$, whose structure describes in turn the five axioms: Hilbert space, self-adjoint observables, tensor product, unitary evolution, and the Born rule.

That the state space can carry this description has its root in three basic facts: excitations are countable, energies add, independent composition multiplies. Counting gives the objects, and merging and independent pairing are the only two ways countable objects combine, the former giving addition and the latter multiplication, so that $(\mathbb{N}^*, +, \times)$ is completely determined. Unique factorization $n = \prod_p p^{a_p}$ holds in it, the energy $H = \log n = \sum_p a_p \log p$ adds prime by prime accordingly, and the primes are the irreducibles under multiplication, the indivisible excitations. The state space is therefore discrete: the dimension is finite, the energy spectrum is discrete, the minimal spacing $\log(1 + 1/n)$ decreases monotonically to zero with n , and for sufficiently large n the macroscopic behavior is indistinguishable from that of a continuous space.

The BC state space has described the five axioms of quantum mechanics. Whether its structure can describe, in finer detail, the physical world made of matter, forces, and spacetime will be the work ahead.

Data Availability

No datasets were generated or analysed in this study; all results are obtained by analytic derivation.

Conflict of Interest

The author declares no conflict of interest.

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